

The Edge Negotiator

Verified-Source Cross-Junction Coordination for SLM-Driven Traffic Signal Control

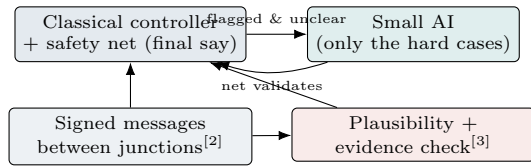
Progress report · UCL MSc Systems Engineering for IoT (25153651) · Supervisors: Dr A. Delibasi (UCL), L. Stott (Microsoft) · 2026-06-20

1. The project and its goal

The system is a corridor of traffic junctions on one arterial. Each junction runs a fast classical controller (*MaxPressure*^[1]) by default, which is also the safety floor. Alongside it runs a small AI model (**Phi-4-mini**, on the junction itself via **Microsoft Foundry Local**, never retrained), called in only for hard cases. Junctions share what they are about to do over a channel that is digitally signed and checked for plausibility.

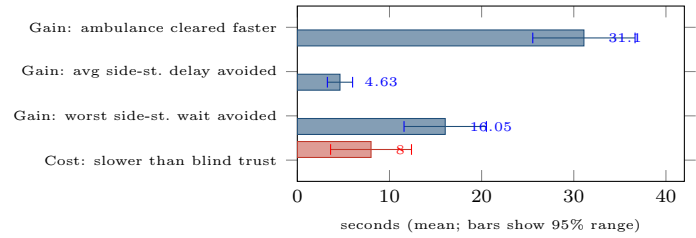
The goal: coordinate well in normal traffic, and fail safely when a neighbour lies. If a junction's messages are tampered with or the junction is hijacked, the system detects that the reports do not match reality and falls back to local control. A deterministic safety net always makes the clear-cut calls: clear an ambulance a junction can physically detect, refuse a claim it cannot verify; safety never depends on the AI. The AI handles only what a fixed rule cannot settle: when an alert is flagged but the evidence is partial or conflicting, it weighs the neighbours' reports to judge a real incident from a spoofed grab. Whether that beats a well-tuned fixed rule is the central question we measure, and we report it either way.

2. How it works



The classical controller runs by default and has the final say on every safety call. Only a flagged, unclear alert goes to the AI. Every message is signed and checked first.

3. Measured effects (30 runs each)



SUMO^[4], 30 paired runs. Gains (blue) vs the no-defence baselines; cost (red) vs blindly trusting. Every 95% range excludes zero.

4. What we have built, and what it shows

- Working and tested:** signed messaging between junctions, an approved-senders registry, the classical controller and its safety net, a check that flags neighbour reports inconsistent with the real traffic, and a four-junction simulated corridor.
- Emergency handling:** a junction clears an ambulance only with physical evidence the vehicle exists; a valid signature is not enough, so a faked emergency is refused.
- Live attack-and-defence demo:** a hijacked junction sends a signed but fake “ambulance approaching” alert and it is refused;
- Evidence (30 runs):** emergency handling clears the ambulance 31s faster than with none; under the spoof attack the worst side-street wait is held 16s below a trust-everything controller; requiring evidence costs 8s of ambulance time versus blindly trusting (a price we accept).
- Independent check:** an automated audit of the write-up found no fabricated sources; a few over-stated citations were corrected. Every result is reproducible from its settings.

5. Where it sits in the literature

To our knowledge, no prior system has been evaluated as a multi-junction, AI-coordinated controller in which a junction's messages can be faked by an insider that still holds a valid key. The parts exist; the combination, tested under that threat, does not.

Work (year)	What it does	Coord?	Trust?	How we differ
CoLLMLight (2025)	Cooperative LLM junctions share state; fine-tuned 8B	Yes	No	Frozen 3.8B; signed + plausibility-checked channel
VLMLight (2025)	Fast-classical + slow-LLM, safety-critical cases	No (1 junction)	No	Multi-junction; small on-device model; compromised-channel test
LA-Light (2024)	LLM calls rule/RL for rare events; cloud GPT-4	Multi (tested)	No	On-device; faults are adversarial and signed, not benign
iLLM-TSC (2024)	LLM refines an RL controller under noisy comms	No	Noise only	Malicious (spoofed) messages, not random noise
LLMLight (2023/25)	Single LLM controls one junction; fine-tuned	No	No	Multi-junction coordination + trust layer
EMVLight (2022)	Trained MARL for emergency-vehicle preemption	Yes	No	No retraining; handles novel incidents; trust layer
REG-TSC (2025)	RAG + fine-tuned 8B agents, emergencies	Yes	No	Frozen 3.8B; RAG optional; signed + checked
CoLight (2019)	Graph-attention RL, cooperative control	Yes	No	Language reasoning on hard cases + security
MaxPressure (2013)	Throughput-optimal classical control	Local	No	Adopted directly as the default and safety floor
SafeLight (2023)	Trained RL + rule-based safety override	No	No	No training; defends against spoofed inputs
Blockchain-TSC (2019)	Blockchain secures vehicle→signal data	No	Yes	Lightweight signed log; secures signal↔signal
proofmember23 (2023)	Device-identity / membership proof (IoT)	No	Yes	Applied to junctions + a content plausibility check
Derhab (2020)	Flow-conservation check for sensor-network attacks	n/a	Yes	Ported to vehicle conservation across junctions
Keijzer (2021)	Theory-only detection of coordination attacks	Yes	Yes	A measured detection-vs-attacker-strength curve; identity + audit

Why this direction. Racing the classical controller on normal traffic is a dead end (it is throughput-optimal and microsecond-fast), so performance is a secondary, honestly-reported axis. The AI earns its place only as a guarded exception handler on hard

cases. The headline is robustness: detecting and degrading safely under an authenticated-but-lying neighbour, which signatures alone cannot catch. We use SUMO (it models emergency vehicles and lane blockage), a signed hash-chained log rather than a heavyweight blockchain (one city authority, one trust domain), and a permissioned key registry rather than full decentralised identity.

6. Value and how we measure success

The deployable angle: a safety-critical demonstration of edge AI (Foundry Local, Phi-4, no cloud, no retraining) with verifiable agent identity, the same trust problem as Entra Agent ID, made physical and measurable. Success is judged on:

Indicator	Target / acceptance
Spoofed-emergency resistance	Every uncorroborated emergency claim refused; no false preemption
Emergency-vehicle delay	Reduced versus no-preemption (measured: 31 s faster over 30 runs)
No-harm guarantee	The security layer does not slow normal traffic beyond a pre-registered margin
Attack-detection quality	Precision / recall / detection-latency reported across a sweep of lie sizes
Edge feasibility	Phi-4-mini decision within budget on Foundry Local; gated so a corridor stays within it
Auditability	Tamper-evident log of every signed message and decision

7. Next steps

- Central experiment: on flagged, unclear alerts, does the AI beat a well-tuned fixed rule? Measure and report either way, including a null.
- Make normal-traffic coordination steer decisions, not only verify them, then measure the green-wave benefit.
- Detection limit: grow the lie until detection fails; report where.
- Real Lambeth arterial and a full demand sweep, with confidence ranges.
- Run the real Phi-4-mini on Foundry Local; confirm repeatable decisions.

8. What we do and do not claim

Claim: authenticated messaging between junctions; detection of faked or faulty reports with a measured reliability curve; a deterministic safety floor; a fair attack-and-defence comparison.

Do not claim: new cryptography; that the AI beats the classical controller on ordinary traffic; that the AI yet improves the ambiguous decisions (measured next); that coordination yet steers normal traffic (verified, not yet causal); defence against colluding insiders or a stolen master key; or that simulator results transfer directly to hardware.

References. [1] Varaiya, Max-pressure control, *Transp. Res. C* 36 (2013). [2] Bernstein et al., Ed25519, *J. Cryptogr. Eng.* (2012). [3] Lighthill & Whitham, *Proc. R. Soc. A* 229 (1955); Richards, *Oper. Res.* 4 (1956) (vehicle-conservation law). [4] Lopez et al., SUMO, *IEEE ITSC* (2018). [5] Yuan et al., CoLLMLight, arXiv:2503.11739 (2025). [6] Wang et al., VLMLight, arXiv:2505.19486 (2025). [7] Wang et al., LA-Light, arXiv:2403.08337 (2024). Also: iLLM-TSC arXiv:2407.06025; LLMLight arXiv:2312.16044; EMVLight arXiv:2206.13441; REG-TSC arXiv:2510.26242; CoLight arXiv:1905.05717; SafeLight arXiv:2211.10871; Blockchain-TSC arXiv:1906.02628; proofmember arXiv:2310.08163; Derhab *Sensors* 20(21):6106 (2020); Keijzer *ECC* (2021).